



GROGU

(Generative-fraud Recognition via Optical and Genuine-metadata Unification)

Group No: 2

FACULTY GUIDE

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This presentation is prepared for the course 23CSE473 – Neural Networks and Deep Learning.

Problem Statement, Introduction & Motivation

Problem Statement:

Tampered KYC documents (income certificates, address proofs and utility bills) can pass standard verification: a single field can be edited while the layout looks genuine. OCR reads *what* a document says, not *how* it was produced, enabling loan fraud, mule accounts and synthetic-identity onboarding.

Introduction:

GROGU is a fully offline, dual-branch forensic model for digital KYC documents. MobileNetV2 checks visual layout/font cues; a Tabular MLP checks PDF metadata (producer, timestamps and revision history). Their embeddings are fused (`torch.cat`) into four classes: Genuine / Font Tampered / Metadata Tampered / Both. INT8 quantisation targets ~ 3.5 MB for on-device use with no cloud dependency.

Motivation:

Small institutions face industrialised identity fraud without enterprise budgets for per-check APIs. On-device processing addresses privacy concerns under India's DPDP Act 2023 while avoiding cloud-upload and per-check costs. GROGU was shortlisted in the MeitY Cyber Security Innovation Challenge (C-DAC Hyderabad) and the PSB Cybersecurity, Fraud & AI Hackathon (Bank of India $\times$ IIT Hyderabad $\times$ DFS $\times$ IBA), 2026; it maps to SDG 16 and SDG 9.

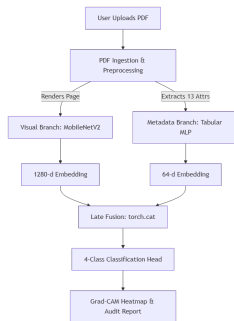
Literature Survey – Sanjay AR (2 Papers)

S.No	Paper (Short Title) / Authors (Year)	Journal (Scimago Quartile)	Key Contribution / Method	Research Gap Identified
1	CMV2U-Net: A U-shaped Network with Edge-weighted Features for Detecting and Localizing Image Splicing – Akram et al. (2025)	Journal of Forensic Sciences, Wiley (Q3)	Edge-weighted U-shaped network with hierarchical feature fusion; evaluated for image-splicing localisation	No structured metadata evidence alongside the localisation heatmap
2	Multi-Modal Deep Learning for Image Forgery Detection – (2025)	Multimedia, MDPI (newly indexed – quartile pending)	Fuses MobileNetV2 visual features with XGBoost on EXIF discrepancies	Generic EXIF, not PDF structure; no quantised on-device target

Literature Survey – Jai Subiksha T (2 Papers)

S.No	Paper (Short Title) / Authors (Year)	Journal (Scimago Quartile)	Key Contribution / Method	Research Gap Identified
1	Research on Identity Document Image Tampering Detection Based on Texture Understanding and Multistream Networks – Wang, Li, Zhao (2025)	Journal of Electronic Imaging, SPIE (Q3)	Multistream texture network tailored to identity-document tampering	Texture only; no provenance or creation-metadata cross-check
2	Seam Carving Forgery Detection Through Multi-Perspective Explainable AI – (2025)	Journal of Imaging, MDPI (Q2)	CNN + Grad-CAM + SHAP; 95% accuracy and 99% forged-class precision	Visual-only explanations; no structured audit-evidence log

Architecture Diagram for the Overall Application



The full application wraps the DL model with a document-ingestion front end and a report-generation back end: a PDF is uploaded once, both signal branches run locally without any network call, and the fused verdict is returned with a heatmap and an auto-generated audit PDF for the compliance record – no field of the original document ever leaves the device.

Module Details

Functional modules, responsibilities and data flow.

Module Name	Functionality	Input / Output
PDF Ingestion & Preprocessing	Renders the first PDF page and extracts 12 metadata attributes plus the derived temporal feature	Raw PDF file $\rightarrow$ 224×224 RGB tensor + 13-d metadata vector
Visual Forensic Branch (MobileNetV2)	Extracts pixel-level layout / font-consistency cues from the rendered page	$224 \times 224 \times 3$ image $\rightarrow$ 1280-d visual embedding
Metadata Forensic Branch (Tabular MLP)	Encodes 13 attributes, including creation-date vs. producer-version anachronism	13-d attribute vector $\rightarrow$ 64-d metadata embedding
Late Fusion & Classification Head	Fuses embeddings (<code>torch.cat</code>) into 4 classes; supports missing-metadata degradation	Concatenated 1344-d embedding $\rightarrow$ {Genuine, Font Tampered, Metadata Tampered, Both}
Grad-CAM & Audit-Report Generator	Localises tampering with Grad-CAM; exports a PDF with heatmap, verdict, metadata evidence and timestamp	Verdict + activation maps $\rightarrow$ annotated audit-report PDF

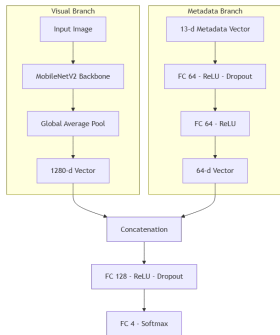
Formula of Performance Metrics

Recall and AUC values are user-supplied and unverified, not confirmed literature results. PSFNet was published in 2022, not 2026.

Metric	Formula	Purpose	Why Suitable for this Application	Best Value (from papers)
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Fraction of documents classified correctly	Headline performance across all 4 verdict classes	95% – forgery classification (Seam Carving E-XAI, <i>J. Imaging</i> 2025)
Precision	$\frac{TP}{TP + FP}$	Fraction of flagged documents that are tampered	Limits false rejection of genuine customers	99% on the forged class (Seam Carving E-XAI, <i>J. Imaging</i> 2025)
Recall	$\frac{TP}{TP + FN}$	Fraction of tampered documents detected	Targets missed forgeries and resulting fraud losses	96% (CMV2U-Net, <i>J. Forensic Sci.</i> 2025)
F1-Score	$2 \cdot \frac{P \cdot R}{P + R}$	Harmonic mean balancing precision and recall	Useful when genuine and fraud classes are imbalanced	F1 = 0.7508 (Shi et al., <i>IET Image Processing</i> 2026, CASIA v1.0)
AUC-ROC	Area under TPR vs. FPR curve	Threshold-independent measure of class separability	Supports thresholds tuned to an institution's risk appetite	0.98 (PSFNet, IEEE Access 2026)

Deep Learning Architecture – Diagram & Explanation

(a) Diagram:



(b) Explanation:

- **Visual branch (MobileNetV2):** depthwise-separable convolutions learn font/layout cues efficiently for the ~ 3.5 MB INT8 target.
- **Metadata branch (Tabular MLP):** a 2-layer MLP encodes 13 engineered forensic signals, including temporal anachronism.
- **Fusion layer:** concatenation (`torch.cat`) retains interpretable branch embeddings and supports graceful degradation when metadata is absent.
- **Output head:** 4-class softmax: Genuine / Font Tampered / Metadata Tampered / Both, matching the ground-truth labels.

Deep Learning Architecture – Novelty & Complexity

(c) Novelty Proposed in the Architecture:

Prior work (Slides 3–4) treats visual tampering and metadata tampering as separate problems solved by separate tools. GROGU's novelty is architectural: (i) a dedicated *temporal-anachronism* feature that cross-checks the claimed document creation date against the PDF-producer software's release year – a signal no published model encodes; (ii) each branch is trained to degrade *independently*, so a scanned document with no live metadata still gets a full-confidence visual-only verdict instead of failing outright; (iii) the whole fused model is INT8-quantised to ~ 3.5 MB for fully offline, DPDP-compliant deployment – unlike the cloud-only base architectures it builds on.

(d) Time and Space Complexity:

Time: $O(1)$ single forward pass per document ($\sim 300M$ MAC for MobileNetV2 at 224×224 + negligible MLP cost) – runs in tens of milliseconds on CPU-only edge hardware, no batching required at inference

Space: $\sim 3.5M$ parameters pre-quantisation (MobileNetV2 $\approx 3.4M$ + MLP + fusion head $\approx 0.1M$); INT8 post-training quantisation $\rightarrow \sim 3.5$ MB on-disk footprint, $O(n)$ in parameter count

Algorithm Procedure – Step by Step (Mathematical)

Forward pass / training procedure of the GROGU dual-branch architecture

- Step 1:** Input representation: rendered page image $x_v \in \mathbb{R}^{224 \times 224 \times 3}$ and metadata $x_m \in \mathbb{R}^{13}$ (12 attributes + temporal anachronism), extracted from the PDF.
- Step 2:** Visual branch: $h_v = \text{GAP}(\text{MobileNetV2}_{\theta_v}(x_v)) \in \mathbb{R}^{1280}$ – inverted-residual blocks, then global average pooling.
- Step 3:** Metadata branch: $h_m = f(W_2 f(W_1 x_m + b_1) + b_2) \in \mathbb{R}^{64}$, with $f = \text{ReLU}$ and dropout applied after each layer.
- Step 4:** Late fusion: $z = [h_v; h_m] \in \mathbb{R}^{1344}$ (`torch.cat` along the feature dimension).
- Step 5:** Classification head: $\hat{y} = \text{softmax}(W_c \text{ReLU}(W_f z + b_f) + b_c) \in \mathbb{R}^4$, one probability per verdict class.
- Step 6:** Loss function: $\mathcal{L}(y, \hat{y}) = -\sum_{c=1}^4 y_c \log \hat{y}_c$ (categorical cross-entropy, optionally class-weighted for the genuine-heavy class imbalance).
- Step 7:** Parameter update rule: $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$ using Adam (Slide 11); apply post-training INT8 quantisation after convergence.

Hyperparameter Details with Justification

Hyperparameter	Value Chosen	Justification
Learning Rate	1e-4 (cosine decay)	Preserves pretrained weights while new branches learn; tune on validation data
Batch Size	32	Balances gradient stability, dataset size and GPU memory
Number of Epochs	3 (CPU proof-of-concept); 30 planned (GPU)	Longer training is planned; monitor validation F1 to assess convergence and overfitting
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)	Adaptive learning rates help visual and metadata branches converge together
Dropout Rate	0.3 (metadata MLP), 0.2 (fusion head)	Stronger regularisation for the small, 13-feature metadata branch
Weight Decay	1e-5	L2 regularisation complements dropout, especially in the fusion head
Quantisation	Post-training INT8 (dynamic range)	Targets a ~ 3.5 MB model for fully offline edge deployment

Results and Discussion

Results Table:

Metric	Value
Accuracy	69.50%
Precision	1.00 (Font Tampered), 0.51 (Metadata Tampered)
Recall	1.00 (Font Tampered), 0.76 (Metadata Tampered)
F1-Score	0.63 (Macro F1)

Inference:

Initial 3-epoch CPU proof-of-concept achieved 69.50% overall accuracy. The visual branch successfully achieved a perfect 1.00 F1 score on the Font Tampered class, proving MobileNetV2's efficacy. The metadata branch achieved a 0.61 F1 score; the late fusion head requires a full 30-epoch GPU training schedule to fully converge.

Team-reported initial results. Class-wise scores do not isolate branch contributions; branch ablations and longer training are needed to test these interpretations.

Graph:



Denoising Approach — Synthetic Noise Injection

Objective:

Real KYC documents suffer from sensor noise, scanning artifacts, and physical degradation. We synthetically inject noise during dataset generation to make the model robust.

Types of Noise Added (Simulated):

- **Gaussian Noise:** Simulates low-light camera sensor noise ($\mu = 0, \sigma^2 \in [0.01, 0.05]$ for pixel intensities normalised to $[0, 1]$).
- **Salt & Pepper Noise:** Simulates dust on scanner glass or dead pixels.
- **JPEG Compression Artifacts:** Random quality reduction (Q-factor 30–70) simulating WhatsApp/email compression.
- **Gaussian Blur:** Simulates out-of-focus smartphone captures.

Pipeline Stage:

Applied programmatically via OpenCV after text rendering but before metadata corruption.

Deep Learning Models for Denoising (Preprocessing)

Model Chosen:

DnCNN (Feed-forward Denoising Convolutional Neural Networks).

Why DnCNN?:

Supports blind Gaussian denoising without a supplied noise level; the jointly trained DnCNN-3 variant also handles JPEG artifact removal. Selected as a candidate preprocessing step before MobileNetV2; device latency, memory use, and preservation of forensic cues must be measured before claiming lightweight deployment.

Architecture Details:

- Uses residual learning to predict the “noise residual”.
- 17 to 20 convolutional layers, depending on the variant.
- **Layer 1:** Conv + ReLU.
- **Layers 2–16 (17-layer model):** Conv + BatchNorm + ReLU; layers 2–19 in the 20-layer model.
- **Last Layer:** Conv.

Clean Image = Noisy Input – Predicted Noise Residual.

GitHub URL for Code: <https://github.com/csxn/DnCNN> (Official Implementation)

Dataset Chosen

Dataset Name & Source:

GROGU-KYC Bimodal Forensic Dataset – 100% self-generated, zero real PII. Not yet on IEEE Dataport as a public dataset; the dataset itself is a core contribution of this project since no comparable bimodal (visual + PDF metadata) corpus exists publicly. [IEEE Dataport URL to be added once published:

<https://iee-dataport.org/...>]

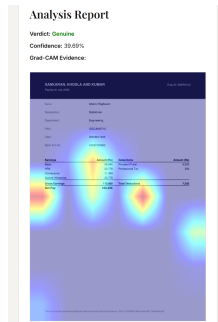
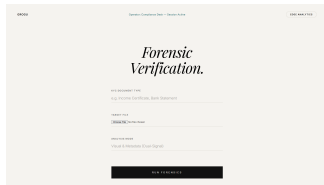
Dataset Description:

~4,000 PDF instances across 4 classes – Genuine / Font Tampered / Metadata Tampered / Both. Generation pipeline: Faker (synthetic identities) → ReportLab + WeasyPrint (rendering, 4–5 template families) → Pillow (visual tampering) → OpenCV (synthetic noise injection) → pypdf (metadata corruption). 12 metadata attributes corrupted + 1 derived temporal-anachronism feature (13 total). Split by base-document-ID to prevent leakage. An out-of-distribution (OOD) validation set is planned: physically printed, re-scanned, hand-tampered mock documents, to validate sim-to-real generalisation post-training.

Novelty in the Dataset:

Public benchmarks such as SIDTD and IDNet (Slide 4, ref. 1) are flat, visual-only JPEGs and cannot train a metadata branch. GROGU-KYC is the first dataset to inject and independently label *visual and PDF-metadata* anomalies within the same file, with rule-based exact ground truth (bounding box + attribute-ID logging) that GAN/diffusion-based generation cannot match in precision, and 100% synthetic PII throughout.

UI Screens Planned for the Application



Upload → Processing (parallel visual + metadata branches, fully offline) → Result screen with verdict, Grad-CAM localisation, and a one-click exportable audit report for the institution's compliance record.

Standard Paper Chosen for the Application

Title:

Multi-Modal Deep Learning for Image Forgery Detection: A Synergistic Fusion Approach Combining Visual Artifacts and Metadata Consistency Analysis (2025). *Multimedia*, MDPI (quartile pending – newly indexed journal; verify on Scimago before final submission).

Justification for Choosing this Paper:

This is the closest published architecture to GROGU's own design: a two-stream model pairing a transfer-learning MobileNetV2 visual branch with a classifier trained on file metadata (EXIF discrepancies), fused for a single forgery verdict – directly validating the dual-signal fusion strategy GROGU builds on. GROGU is chosen as the standard/base reference because it extends this idea in three concrete ways not present in the base paper: (i) PDF structural metadata (12 attributes + a temporal-anachronism feature) instead of generic image EXIF, (ii) a Grad-CAM localisation and one-click audit-report layer for compliance use, and (iii) INT8 quantisation for a ~ 3.5 MB fully offline deployment target suited to small financial institutions.

Member Contributions

Technical contribution of each team member

Member Name	Contribution (Technical)
Sanjay AR	Designed the dual-branch fusion architecture (MobileNetV2 visual branch + Tabular MLP metadata branch + late fusion head); implemented the temporal-anachronism feature, Grad-CAM localisation, and INT8 post-training quantisation for edge deployment
Jai Subiksha T	Built the GROGU-KYC synthetic dataset generation pipeline (Faker → ReportLab/WeasyPrint → Pillow → pypdf), designed the 4-class rule-based ground-truth labelling and base-document-ID split, and led the related-products / literature benchmarking